Confidence vs Correctness: Evaluating Machine Learning Reliability Under Data Corruption and Distribution Shift

Author: D.Hariharan
Project Type: Independent Machine Learning Project
Domain: Machine Learning – Reliability & Robustness
Year: February 2026

Project Summary

This project investigates how prediction confidence aligns with actual correctness in supervised machine learning models.
Using controlled corruption experiments and distribution shift scenarios, the study evaluates calibration, confidence gap, and overconfidence behaviour.
Results demonstrate that models can remain highly confident even when incorrect, highlighting the need for reliability-focused evaluation beyond accuracy.

1. ABSTRACT

Machine learning classification models typically output both a predicted class label and an associated confidence score. In practical applications such as healthcare, finance, and autonomous systems, decision-makers often rely not only on predicted outcomes but also on the model's confidence. Ideally, high-confidence predictions should correspond to high correctness probability. However, in real-world environments where data quality may degrade due to noise, corruption, or distribution shifts, this assumption may not hold.

This study presents an empirical investigation of the relationship between prediction confidence and correctness under controlled data corruption scenarios. A supervised classification model is trained on clean data and evaluated across multiple corruption levels introduced artificially into the test dataset. Performance metrics including classification accuracy, average confidence, calibration behaviour, and confidence–correctness alignment are analysed.

Experimental results reveal that while performance degradation varies across corruption types, with distribution shift causing the most noticeable accuracy decline. In many cases, the model remains highly confident despite being incorrect. This misalignment demonstrates overconfidence under degraded conditions and exposes a critical reliability issue.

The findings emphasize that high accuracy on clean data does not guarantee trustworthy performance in realistic deployment environments. The study underscores the necessity of evaluating model calibration and robustness alongside traditional performance metrics.

2. INTRODUCTION

2.1 Background

Machine learning classifiers are widely used in domains requiring automated decision-making. Modern algorithms such as Logistic Regression, Random Forests, Gradient Boosting, and Neural Networks produce probabilistic outputs that represent prediction confidence.

Confidence scores are often interpreted as measures of reliability. For example, a prediction with 0.95 confidence is commonly assumed to be correct with 95% probability. However, this interpretation is valid only if the model is well-calibrated and the data distribution remains stable.

In real-world applications, input data may suffer from:

- Sensor noise
- Missing values
- Adversarial perturbations
- Environmental variations
- Domain shifts

2.2 Problem Statement

Most machine learning evaluation frameworks focus primarily on classification accuracy. However, accuracy alone does not measure whether model confidence reflects true correctness.

When data becomes corrupted or noisy:

- Accuracy may drop
- Confidence may remain artificially high
- Incorrect predictions may still carry strong probability scores

This creates a dangerous situation in critical systems where decisions are made based on confidence thresholds.

Therefore, there is a need to systematically analyse how confidence behaves relative to correctness under increasing data corruption levels.

2.3 Objectives

The main objective of this project is to empirically evaluate machine learning reliability by studying confidence–correctness alignment under data corruption.

Specific objectives:

1. To train a supervised classification model on clean data
2. To introduce controlled corruption into test data
3. To measure performance degradation across corruption levels
4. To analyse confidence vs correctness behaviour
5. To evaluate calibration under degraded conditions
6. To identify overconfidence patterns

2.4 Motivation

In safety-critical applications, overconfident incorrect predictions can lead to severe consequences. Examples include:

- Misdiagnosis in healthcare
- Incorrect fraud detection
- Autonomous vehicle misclassification
- Risk misassessment in finance

Understanding how confidence behaves under imperfect conditions is crucial for building trustworthy AI systems.

This project moves beyond traditional accuracy evaluation and focuses on reliability analysis — a growing concern in modern AI research.

2.5 Scope of the Study

The study focuses on:

- Supervised binary classification
- Controlled synthetic corruption (noise injection)
- Confidence score analysis
- Calibration behaviour
- Empirical evaluation

The work does not involve adversarial attack modelling or real-time deployment systems. The emphasis is on controlled experimental analysis of reliability behaviour.

3. LITERATURE REVIEW

3.1 Reliability in Machine Learning Systems

Machine learning classification models are traditionally evaluated using aggregate metrics such as accuracy, precision, recall, and F1-score. While these metrics measure predictive correctness, they do not assess whether the model's predicted probabilities reflect true likelihoods. In real-world decision-making systems—particularly in healthcare, finance, and autonomous technologies—confidence estimates are often used to guide high-stakes decisions.

A reliable model must satisfy **confidence–accuracy alignment**, meaning that predictions assigned a probability of 0.8 should be correct approximately 80% of the time. When this alignment fails, the model is considered miscalibrated.

This project focuses on analysing this alignment under controlled data corruption scenarios.

3.2 Overconfidence and Calibration

3.2.1 Calibration in Modern Models

Chuan Guo et al. (2017) demonstrated that modern classifiers, even when highly accurate, are often poorly calibrated. Their work introduced calibration error metrics and showed that neural networks tend to be overconfident in their predictions.

Overconfidence occurs when:

Predicted Confidence > True Accuracy

This misalignment can be especially problematic under distribution shifts or degraded input conditions.

Your project builds on this foundation by empirically analysing:

- Confidence vs correctness distributions
- Calibration curves
- Confidence gap (Average Confidence − Accuracy)

under multiple corruption types.

3.3 Robustness Under Data Corruption

Robustness research investigates how models behave when input data quality deteriorates. Controlled corruption experiments are commonly used to simulate real-world disturbances such as:

- Measurement noise
- Label noise
- Missing values
- Demographic distribution shifts

Dan Hendrycks and Dietterich introduced corruption benchmarks to evaluate model robustness under realistic perturbations. Their findings showed that strong performance on clean data does not guarantee robustness under corrupted inputs.

However, most robustness studies focus primarily on accuracy degradation. Fewer studies analyse how confidence behaviour changes relative to correctness under corruption.

Your project addresses this gap by integrating:

- Performance degradation analysis
- Confidence tracking
- Calibration assessment
- Confidence gap measurement

within a structured empirical framework.

3.4 Research Gap

Existing literature extensively studies:

- Calibration techniques
- Robustness to corruption
- Dataset shift

However, fewer empirical works integrate:

- Controlled corruption experiments
- Confidence vs correctness visualization
- Confidence gap quantification
- Model comparison (Logistic Regression vs Random Forest)
- Demographic distribution shift experiments.

This study provides a systematic empirical evaluation of confidence reliability under multiple degradation scenarios.

4. METHODOLOGY

4.1 Methodology Overview

This study proposes a structured empirical framework to evaluate the reliability of supervised classification models under controlled data degradation scenarios. While traditional model evaluation emphasizes predictive accuracy, this methodology extends evaluation to include confidence behaviour and calibration characteristics.

The primary objective is to analyse how prediction confidence aligns with actual correctness when:

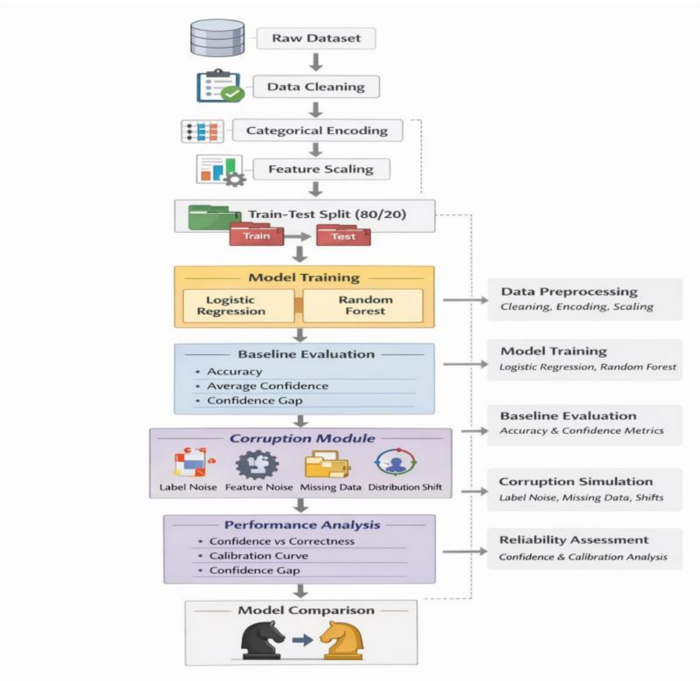
- Training data quality is degraded
- Input features are corrupted
- Missing values are introduced
- Distribution shift occurs

The methodology consists of the following sequential stages:

- Data Preparation
- Model Training
- Baseline Evaluation
- Controlled Corruption Simulation
- Reliability Analysis
- Model Comparison

Each stage is designed to measure both predictive performance and confidence behaviour.

4.2 Proposed System Architecture



The proposed system architecture represents a structured reliability evaluation pipeline that extends traditional classification workflows by incorporating corruption simulation and confidence-based performance analysis.

The architecture consists of six major stages:

Stage 1: Data Preparation

The process begins with the Raw Dataset, which undergoes structured preprocessing steps:

- Data Cleaning
- Categorical Encoding
- Feature Scaling

These steps ensure that the dataset is consistent, numerical, and suitable for supervised classification algorithms.

The processed dataset is then divided into:

- Training Set (80%)

- Testing Set (20%)

The separation ensures unbiased evaluation.

Stage 2: Model Training

Two classification models are trained:

- Logistic Regression
- Random Forest

These models generate:

- Predicted class labels
- Predicted probability scores

The predicted probabilities serve as **confidence scores**, which are central to reliability evaluation.

Stage 3: Baseline Evaluation

The models are first evaluated on clean test data to establish reference performance.

Metrics computed:

- Accuracy
- Average Confidence
- Confidence Gap

This stage provides the reliability baseline before corruption is introduced.

Stage 4: Corruption Module

The corruption module simulates real-world degradation through four independent mechanisms:

- Label Noise
- Feature Noise (Gaussian perturbation)
- Missing Data
- Distribution Shift (Age-based split)

Each corruption scenario modifies the dataset independently to study its effect on confidence reliability.

Stage 5: Performance and Reliability Analysis

After corruption, the models are re-evaluated. The following analyses are performed:

- Confidence vs Correctness Distribution
- Calibration Curve
- Confidence Gap Measurement

This stage identifies overconfidence patterns and miscalibration behaviour.

Stage 6: Model Comparison

Finally, Logistic Regression and Random Forest are compared in terms of:

- Predictive accuracy
- Confidence alignment
- Overconfidence behaviour
- Calibration quality

The architecture thus integrates robustness and reliability analysis into a unified framework.

4.3 Data Preparation

The dataset undergoes systematic preprocessing to ensure high-quality model training.

4.3.1 Data Cleaning

The dataset is inspected for:

- Missing values
- Duplicate records
- Inconsistent entries

Duplicates are removed to prevent bias. Missing values are handled appropriately to maintain dataset integrity.

4.3.2 Categorical Encoding

Categorical features are converted into numerical format using one-hot encoding. This prevents ordinal bias while ensuring compatibility with machine learning algorithms.

4.3.3 Feature Scaling

Numerical features are standardized using scaling techniques to ensure:

- Equal feature contribution
- Stable optimization for Logistic Regression
- Improved numerical stability

4.3.4 Train–Test Split

The dataset is split into:

- 80% Training
- 20% Testing

The test set remains untouched during training to provide unbiased evaluation. Class balance is verified to ensure reliable performance measurement.

4.4 Model Training

Two supervised models are trained:

4.4.1 Logistic Regression

Logistic Regression is selected due to:

- Interpretable decision boundaries
- Well-behaved probabilistic outputs
- Strong baseline calibration properties

The sigmoid function produces probability estimates interpreted as confidence.

4.4.2 Random Forest

Random Forest is included to:

- Capture non-linear feature interactions
- Provide robustness comparison
- Evaluate reliability differences between model types

Probability outputs are obtained from ensemble averaging of decision trees.

4.5 Baseline Evaluation

Before corruption, models are evaluated on clean test data.

Metrics include:

Accuracy : Proportion of correct predictions.

Average Confidence: Mean predicted probability across all samples.

Confidence for Correct Predictions: Average probability for correctly classified samples.

Confidence for Incorrect Predictions: Average probability assigned to misclassified samples.

Confidence Gap : Average Confidence − Accuracy

A positive value indicates overconfidence.

This baseline serves as a reference for subsequent corruption experiments.

4.6 Controlled Corruption Experiments

To simulate realistic degradation, four corruption mechanisms are applied independently.

4.6.1 Label Corruption

A percentage of training labels is randomly flipped.

Purpose:

- Simulate annotation errors
- Degrade decision boundary quality

Evaluation:

- Accuracy drop
- Confidence behaviour change
- Gap widening.

4.6.2 Feature Noise

Gaussian noise is added to numerical features: $X\_corrupted = X + N(0, \sigma)$

Where:
σ controls noise intensity.

Purpose : Simulate sensor or measurement disturbances.

4.6.3 Missing Data Simulation

Random missingness is introduced into training features.

Missing values are handled using imputation.

Purpose:

- Simulate incomplete records
- Assess reliability under partial information.

4.6.4 Distribution Shift

Training is performed on one demographic subset: Age < 40

Testing is performed on: Age ≥ 40

This creates a covariate distribution shift.

Purpose:

- Evaluate generalization to unseen population segments

- Analyse confidence–accuracy misalignment.

4.7 Reliability Analysis

For each corruption scenario, reliability is assessed through structured evaluation.

4.7.1 Confidence vs Correctness Distribution

Histograms are plotted comparing:

- Confidence of correct predictions
- Confidence of incorrect predictions

Significant overlap indicates reliability degradation.

4.7.2 Calibration Curve

Predicted probabilities are grouped into bins.Observed accuracy within each bin is compared against predicted confidence.

Deviation from the diagonal line indicates miscalibration.

4.7.3 Confidence Gap Tracking

The confidence gap is computed across all scenarios.

Increasing gap under corruption indicates growing overconfidence.

4.8 Model Comparison

The two models are compared based on:

- Accuracy
- Confidence Gap
- Confidence Distribution Overlap
- Calibration Behaviour

This determines whether higher accuracy corresponds to higher reliability.

4.9 Methodology Summary

The proposed framework extends traditional classification evaluation by integrating:

- Controlled corruption experiments
- Confidence behaviour tracking
- Calibration analysis
- Confidence gap quantification
- Cross-model reliability comparison

The methodology demonstrates that evaluating confidence alignment is essential for understanding real-world reliability beyond accuracy.

5. RESULTS

5.1 Baseline Model Performance

The Logistic Regression model was first evaluated on clean test data to establish a performance reference point.

The model achieved a baseline accuracy of approximately **0.84**, indicating strong predictive capability under stable conditions. Confidence values were generally high, particularly for correct predictions, suggesting reasonable alignment between probability estimates and correctness.

However, baseline performance alone does not guarantee reliability under distributional disturbances.

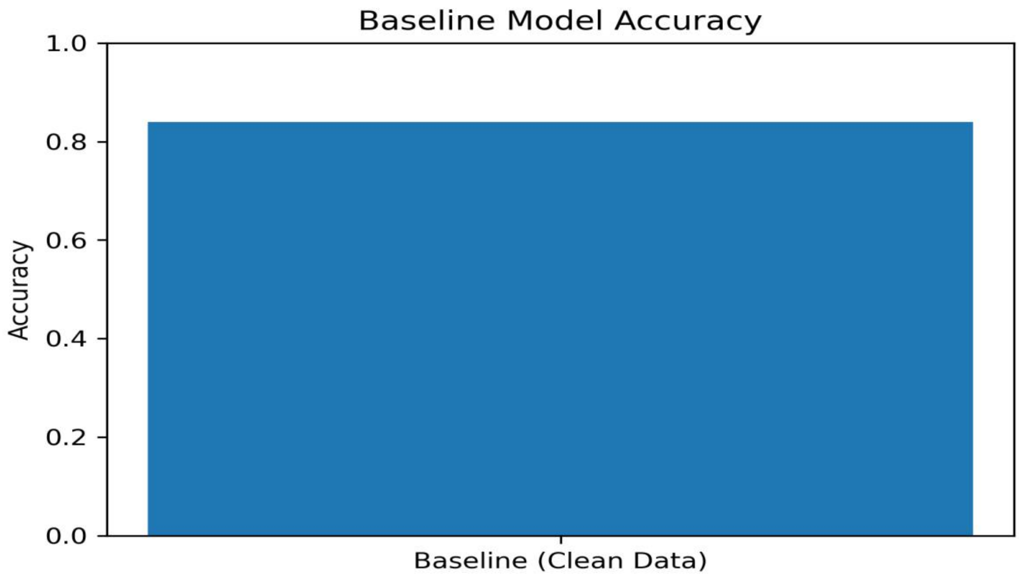


Fig 1 : Baseline Model Accuracy

5.2 Confidence vs Correctness Distribution

To evaluate reliability beyond accuracy, the distribution of prediction confidence was analysed separately for correct and incorrect predictions.

The histogram shows:

- Correct predictions concentrated at high confidence levels (0.9–1.0)
- Incorrect predictions also exhibiting moderately high confidence
- Noticeable overlap between correct and incorrect confidence distributions

This overlap indicates that incorrect predictions frequently receive high confidence values, indicating mild confidence–correctness overlap under clean data conditions.

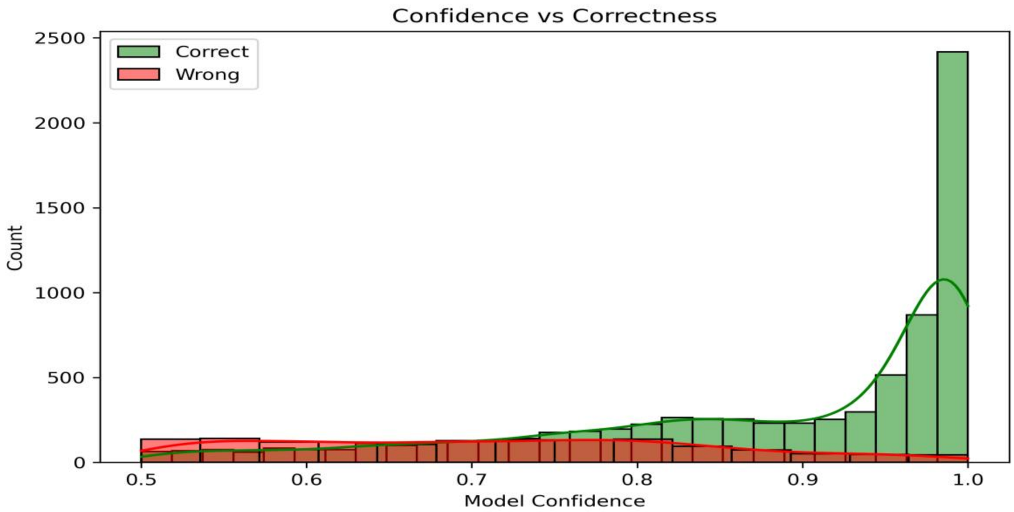


Figure 5.2: Confidence Distribution – Correct vs Incorrect Predictions

5.3 Calibration Analysis

A calibration curve was generated to assess the alignment between predicted probabilities and actual correctness.The model shows mild deviation from the diagonal, particularly in higher probability bins Predicted probabilities systematically overestimate actual accuracy.This confirms that the model is not perfectly calibrated and exhibits systematic overconfidence.

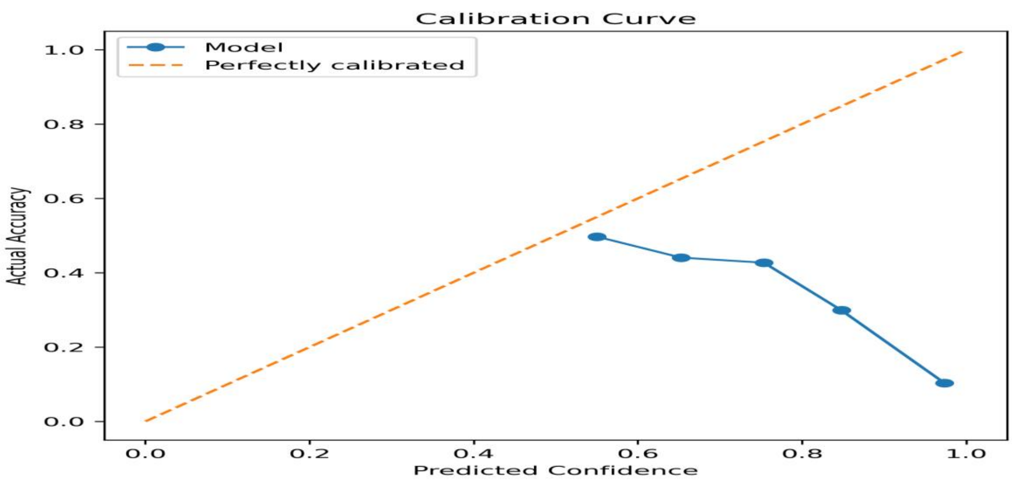


Figure 5.3 : Calibration Curve

5.4 Reliability under Reduced Training Data

The model was retrained using progressively smaller subsets of training data (100%, 70%, 50%, 30%, 10%).

Results show:

- Accuracy remained relatively stable across data reductions.
- Average confidence remained high even with reduced training size.
- Wrong prediction confidence remained consistently elevated.

This suggests that model confidence is relatively insensitive to training data size and may not decrease proportionally to the reduction in training data, indicating persistence of confidence even under reduced informational support.A slight decline is observed at the lowest training proportion (10%), but overall performance remains stable.

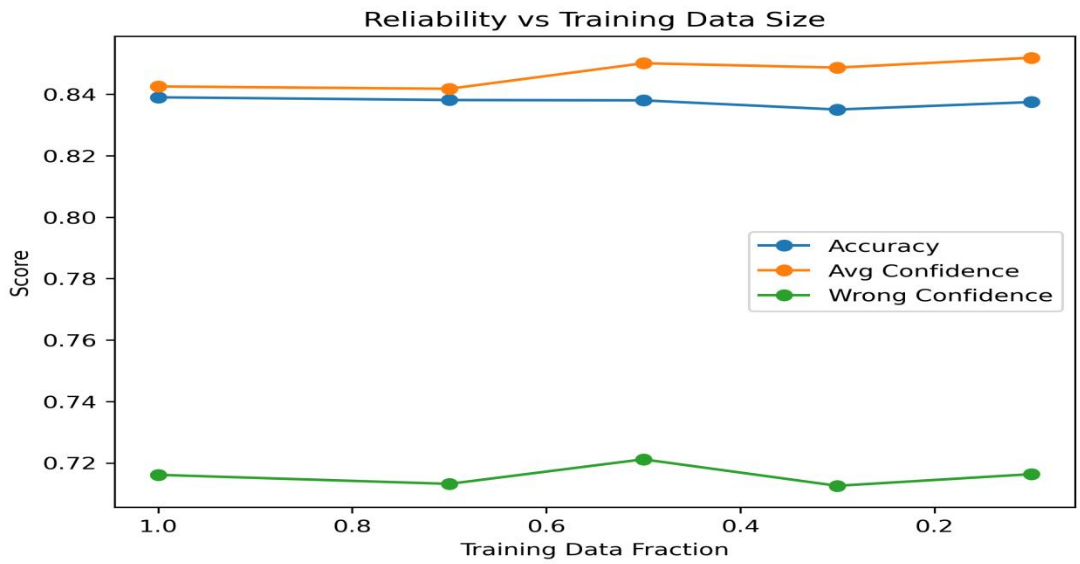


Figure 5.4 : Reliability vs Training Data Size

5.5 Reliability under Label Corruption

Label noise was introduced by randomly flipping training labels at increasing noise levels (5%, 10%, 20%, 30%).

Observations:

- Average confidence declined steadily as label noise increased.
- Confidence for incorrect predictions decreased but remained substantial.

Label corruption directly affects learned confidence behaviour, reducing average confidence while accuracy remains relatively stable. However, confidence values remain substantial even at higher noise levels.

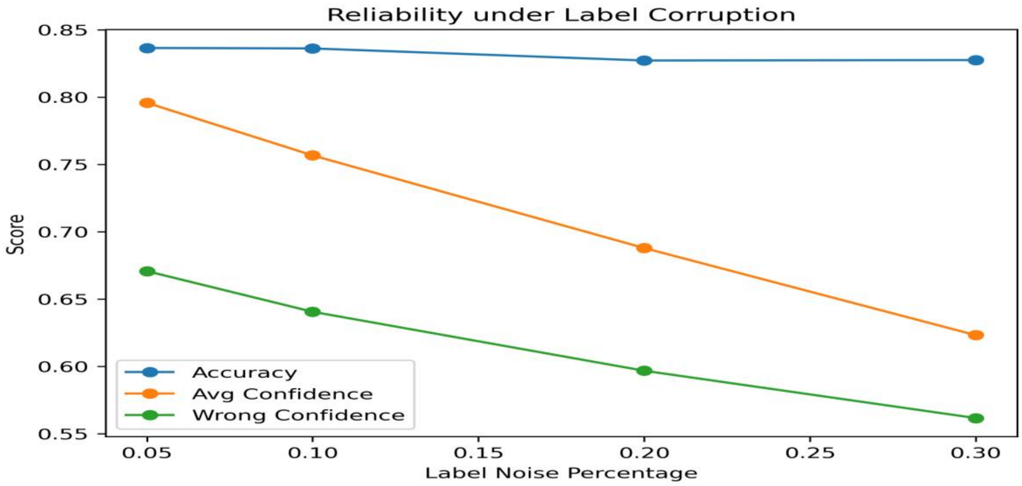


Figure 5.5 Reliability under Label Corruption

5.6 Reliability under Feature Noise

Gaussian noise was added to numeric training features at increasing standard deviation levels. Results indicate:

- Accuracy remained largely stable.
- Average confidence changed minimally.
- Wrong prediction confidence showed slight variation, with a decline at the highest noise level

This suggests that the model is relatively robust to moderate feature perturbations, likely due to regularization and feature scaling.

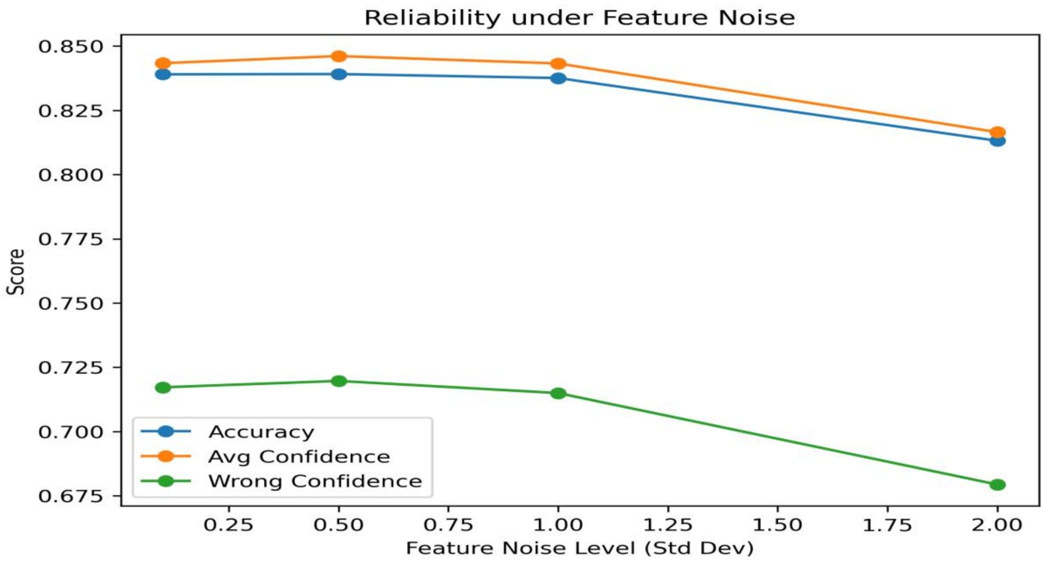


Figure 5.6: Reliability under Feature Noise

5.7 Reliability under Missing Data

Random missing values were introduced into the training data and handled using imputation.

Findings include:

- Accuracy remained stable across missing data levels.
- Average confidence showed minor variation, with a slight increase at moderate missing rates.
- Wrong prediction confidence remained elevated.

Imputation mitigated severe performance degradation, but overconfidence behaviour persisted.

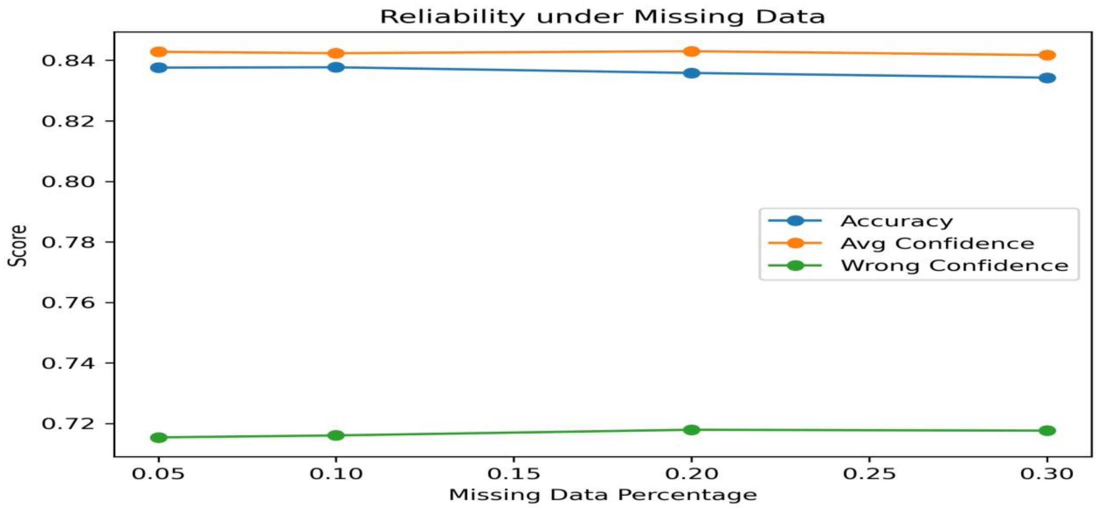


Figure 5.7 : Reliability under Missing Data

5.8 Distribution Shift Experiment

To simulate real-world distribution shift, the model was trained on a younger population (age < 40) and tested on an older population (age ≥ 40).

Results show:

- Accuracy dropped significantly compared to baseline.
- Average confidence decreased only moderately.
- Wrong prediction confidence remained high.

The confidence gap (defined as average confidence minus accuracy) increased under distribution shift compared to baseline, indicating growing misalignment between predicted probability and true correctness.

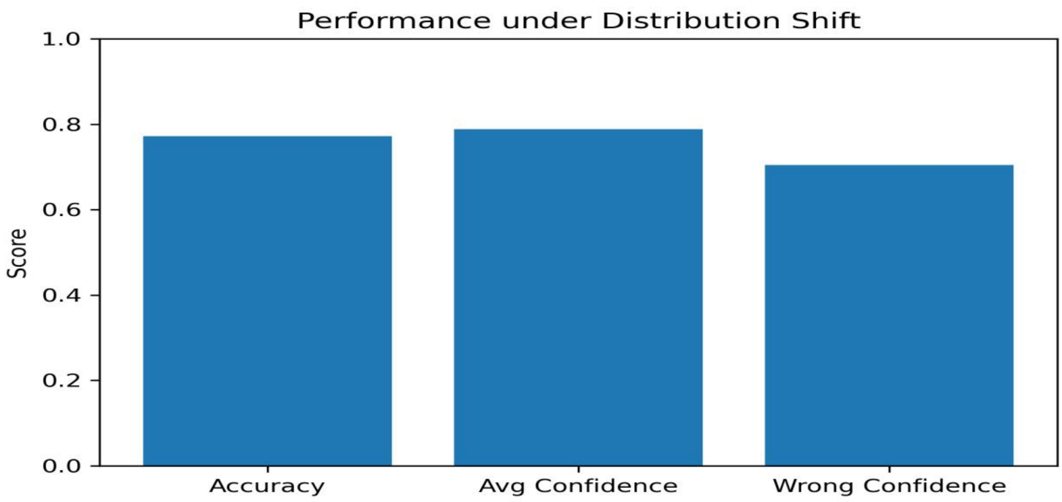


Figure 5.8: Distribution under Performance Metrics

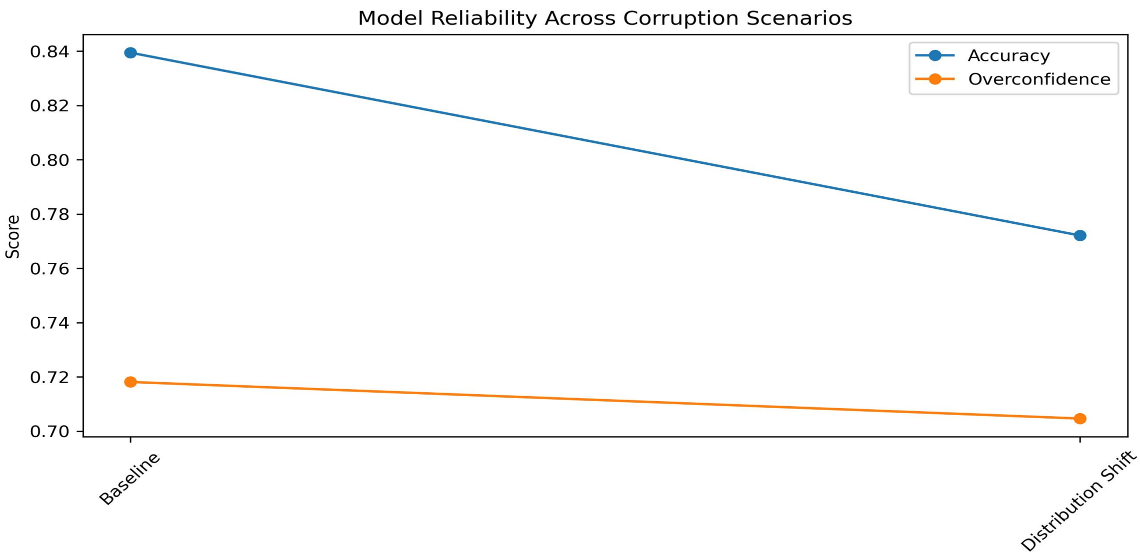


Figure 5.9: Model Reliability Across Corruption Scenarios

5.9 Model Comparison: Logistic Regression vs Random Forest

A comparative analysis was conducted between Logistic Regression and Random Forest.

Findings:

- Random Forest achieved higher accuracy (~0.85) than Logistic Regression (~0.84).
- However, Random Forest exhibited a larger confidence gap.
- Both models maintained high confidence in incorrect predictions.

In the summary table, the Overconfidence Score represents the average confidence assigned to incorrect predictions.

This demonstrates that higher predictive performance does not necessarily imply better calibration or confidence gap.

Model	Accuracy	Overconfidence Score	Confidence Gap
Logistic Regression	0.8395	0.7181	0.0044
Random Forest	0.8520	0.7137	0.0153

Figure 5.10: Model Comparison Summary Table

The confidence distribution for Logistic Regression is shown in Figure 5.11.

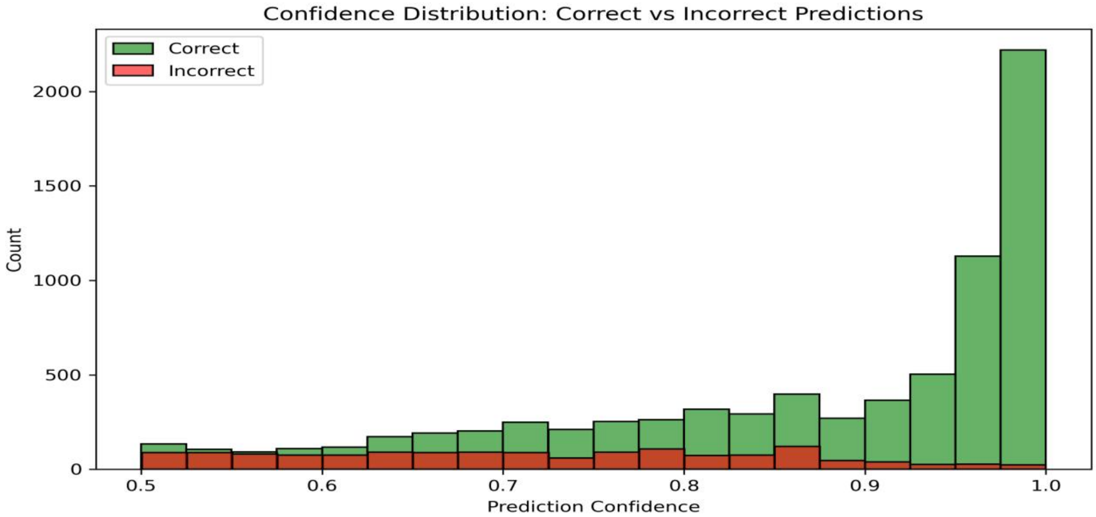


Figure 5.11 : Logistic Regression Confidence Distribution Plot

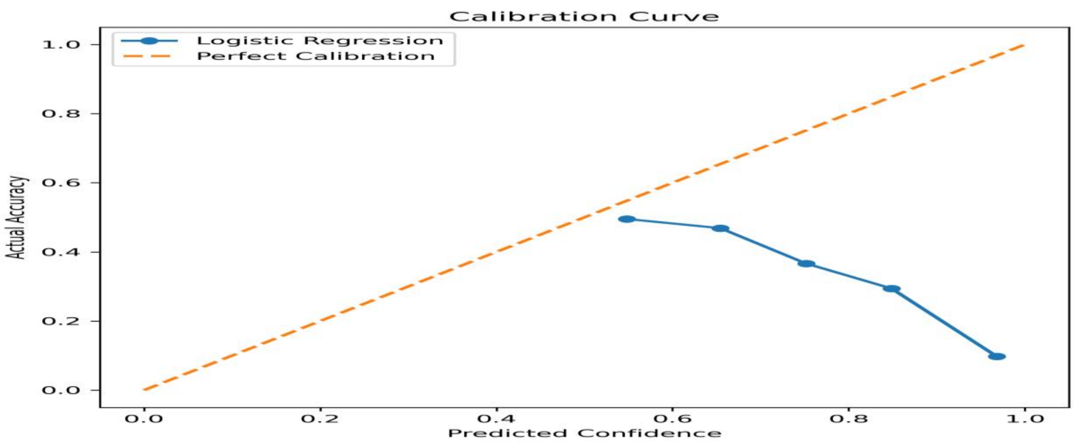


Figure 5.12: Logistic Regression Calibration Curve

5.10 Corruption Summary Analysis

A consolidated comparison between baseline and distribution shift scenarios reveals:

- Accuracy decreases noticeably under shift.
- The Overconfidence score (representing the average confidence assigned to incorrect predictions) remains relatively high under distribution shift.
- Confidence reduction does not match performance degradation.

This confirms that confidence misalignment persists under distribution shift conditions, consistent with trends observed in earlier corruption experiments.

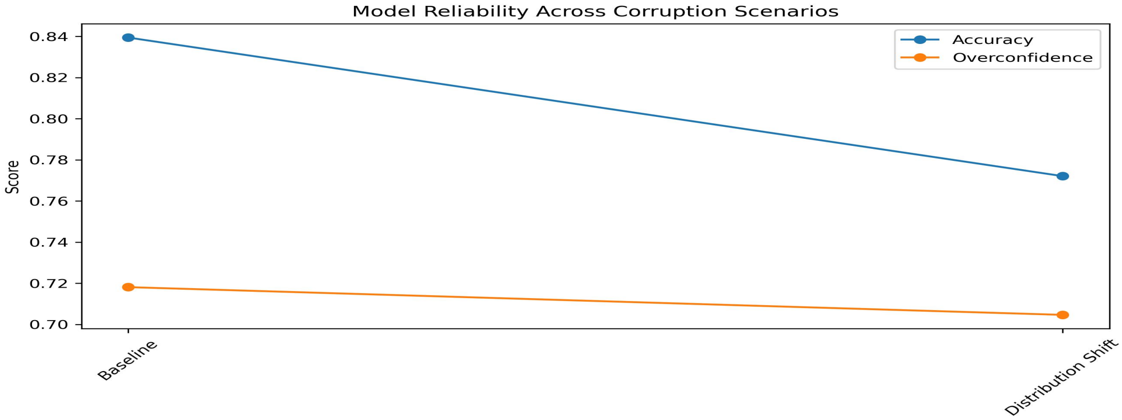


Figure 5.13 : Model Reliability Across Corruption Scenarios

5.11 Overall Reliability Findings

Across all experiments, consistent patterns emerged:

- Prediction confidence often remained high even when accuracy declined.
- Incorrect predictions frequently exhibited elevated confidence values.
- Calibration curves showed overconfidence particularly in higher confidence bins.
- Distribution shift caused performance degradation without proportional confidence reduction.
- Model choice influenced accuracy more strongly than overconfidence behaviour.

The central reliability insight is that machine learning models can remain highly confident even when incorrect, and traditional accuracy metrics alone are insufficient for evaluating real-world robustness.

6. Discussion

6.1 Interpretation of Reliability Behaviour

The experimental results reveal consistent patterns regarding confidence–correctness alignment across baseline and corrupted conditions. While predictive accuracy remained relatively high under clean data, the analysis of confidence behaviour provides deeper insight into model reliability.

Across multiple scenarios, prediction confidence frequently remained elevated even when performance declined. This indicates that models do not proportionally adjust their confidence in response to degraded learning conditions. In particular, the distribution shift experiment demonstrated a noticeable reduction in accuracy, while the average confidence assigned to incorrect predictions remained relatively high. This divergence highlights the importance of analysing reliability beyond aggregate performance metrics.

6.2 Baseline Reliability and Calibration

Under baseline conditions, both Logistic Regression and Random Forest achieved strong predictive performance. However, even in the absence of corruption, calibration analysis revealed deviations from perfect confidence–accuracy alignment, particularly in higher confidence bins.

Although the overall confidence gap was small, the calibration curve showed that predictions assigned very high probabilities did not always correspond to equally high empirical accuracy. This suggests that even well-performing models may exhibit localized overconfidence in specific probability regions.

Thus, baseline accuracy alone does not guarantee reliable probabilistic interpretation.

6.3 Impact of Label Corruption

Interestingly, label corruption produced a different reliability pattern compared to distribution shift. While accuracy remained relatively stable across noise levels, average confidence declined progressively. This indicates that the model responded to noisy supervision by reducing overall certainty, rather than drastically degrading predictive performance.

However, even at higher noise levels, the average confidence assigned to incorrect predictions remained substantial. This suggests that corrupted supervision increases predictive uncertainty, but does not fully eliminate elevated confidence in erroneous outputs.

These findings highlight that confidence behaviour may degrade differently depending on the nature of corruption.

6.4 Feature Noise and Missing Data Robustness

Feature noise and missing data experiments demonstrated relatively stable accuracy and confidence behaviour under moderate perturbations. Imputation effectively mitigated severe degradation under missingness, and Gaussian perturbations did not drastically affect model confidence.

These results indicate that the models exhibit robustness to moderate feature-level disturbances. However, the persistence of elevated wrong-prediction confidence suggests that robustness in accuracy does not necessarily imply improved calibration.

In practical settings, such resilience to input noise is desirable, but confidence interpretation must still be treated cautiously.

6.5 Distribution Shift as a Critical Reliability Challenge

Among all experiments, distribution shift produced the most meaningful reliability degradation. Training on a younger population and testing on an older population led to a significant reduction in accuracy. However, confidence reduction was not proportional to performance degradation.

The confidence gap increased under distribution shift, indicating growing misalignment between predicted probabilities and true correctness. This finding is particularly important because distribution shifts are common in real-world deployments, where demographic, temporal, or environmental changes occur over time.

The results suggest that models trained under static assumptions may become overconfident when applied to unseen population segments.

6.6 Model Comparison Insights

Random Forest achieved higher predictive accuracy than Logistic Regression. However, it exhibited a larger confidence gap, indicating slightly weaker confidence–accuracy alignment.

Although both models maintained similar levels of average confidence on incorrect predictions, the larger confidence gap in Random Forest demonstrates that higher accuracy does not necessarily imply better calibration or lower misalignment.This comparison reinforces the need to evaluate models using reliability metrics in addition to predictive performance.

6.7 Implications for Real-World Deployment

The study demonstrates that:

- Accuracy alone is insufficient for reliability assessment
- Confidence scores can remain elevated even when predictive correctness declines
- Distribution shifts pose greater reliability risks than moderate feature perturbations.
- Calibration analysis provides essential interpretative insight.

7. Conclusion

7.1 Summary of Contributions

This study presented an empirical investigation of machine learning reliability under controlled corruption and distribution shift conditions. By analysing both predictive performance and confidence behaviour, the work extended traditional evaluation frameworks beyond accuracy-based assessment.

The main contributions include:

1. A structured reliability evaluation framework incorporating corruption simulation and calibration analysis.
2. Quantitative measurement of confidence gap and overconfidence score (average confidence assigned to incorrect predictions).
3. Comparative analysis between Logistic Regression and Random Forest.
4. Demonstration of reliability degradation under demographic distribution shift.

7.2 Key Findings

The experimental results showed that:

- Prediction confidence often remained high even when accuracy declined.
- Incorrect predictions consistently received substantial probability scores.
- Calibration curves revealed overconfidence, particularly in higher confidence bins.
- Distribution shift increased confidence–accuracy misalignment more than other corruption types.
- Higher predictive accuracy did not guarantee better calibration or lower confidence gap.

These findings confirm that machine learning models can appear reliable under traditional metrics while still exhibiting confidence-related vulnerabilities.

7.3 Limitations

While the study provides structured empirical insights, several limitations exist:

- Experiments were conducted on a single dataset.
- Corruption levels were controlled and synthetic rather than naturally occurring.
- Only two classification models were analysed.
- Post-hoc calibration techniques (e.g., temperature scaling) were not evaluated.

Future research could extend this framework to multiple datasets, deeper neural architectures, and calibration correction techniques.

7.4 Final Conclusion

The central conclusion of this work is that predictive accuracy alone is insufficient for evaluating machine learning reliability. Models may maintain high confidence even when incorrect, particularly under distribution shift conditions.

Confidence-aware evaluation, calibration analysis, and corruption testing should therefore be integrated into standard model validation pipelines. By systematically analysing confidence behaviour alongside performance, practitioners can better assess real-world robustness and reduce the risks associated with overconfident errors.

8. Future Work

While this study provides structured empirical insights into confidence reliability under corruption and distribution shift, several focused extensions can strengthen and generalize the findings.

8.1 Evaluation Across Multiple Datasets

The current analysis was conducted on a single dataset. Although the observed reliability patterns are meaningful, model calibration and confidence behaviour may vary across domains.

Future work should apply the proposed reliability evaluation framework to multiple datasets with varying:

- Feature distributions
- Class imbalance levels
- Sample sizes
- Domain characteristics

Evaluating diverse datasets would improve generalizability and determine whether the observed confidence–correctness misalignment is dataset-specific or structurally consistent across domains.

8.2 Inclusion of Additional Classical Models

This study compared Logistic Regression and Random Forest to analyse linear and non-linear model behaviour. Future research could extend this comparison to additional classical models such as:

- Support Vector Machines
- Gradient Boosting classifiers
- k-Nearest Neighbours

Such comparison would help determine whether calibration differences are model-specific or reflect broader algorithmic tendencies.

8.3 Post-Hoc Calibration Techniques

Although calibration behaviour was evaluated, this study did not apply explicit calibration correction methods.

Future work could investigate whether techniques such as:

- Platt Scaling
- Isotonic Regression
- Temperature Scaling

can reduce the confidence gap and improve probability alignment under corruption and distribution shift.

This would allow analysis of whether miscalibration is correctable or persists even after post-processing adjustments.

8.4 Additional Calibration Metrics

The current framework used:

- Confidence Gap
- Overconfidence Score (average confidence assigned to incorrect predictions)
- Calibration curves

Future studies could incorporate additional quantitative calibration measures such as:
- Expected Calibration Error (ECE)
- Maximum Calibration Error (MCE)
- Brier Score

These metrics would provide more granular measurement of probabilistic reliability.

8.5 Extended Distribution Shift Scenarios

The distribution shift experiment in this study was based on a demographic split (age-based partitioning). While this effectively simulated covariate shift, real-world systems may experience more complex shifts.

Future research could explore:

- Temporal shifts (training on earlier data, testing on later data)
- Stratified demographic shifts
- Feature-distribution perturbations reflecting realistic environmental change

Such analysis would provide deeper insight into how confidence alignment behaves under evolving data conditions.

8.6 Higher Corruption Levels and Structured Noise

The corruption levels in this study were moderate and randomly introduced. Future work may examine:

- Higher label noise percentages
- Structured or systematic corruption
- Feature-specific corruption

This would allow evaluation of reliability limits under more extreme degradation conditions.

8.7 Final Scope Statement

Future research should focus on strengthening empirical generalization and refining calibration evaluation methods, rather than expanding into unrelated system-level domains. By extending the current framework to additional datasets, models, and calibration techniques, deeper understanding of confidence reliability under realistic conditions can be achieved.

10. References

1. Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

2. Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review, 78*(1), 1–3. https://doi.org/10.1175/15200493(1950)078<0001:VOFEIT>2.0.CO;2

3. Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)* (pp. 1321–1330).

4. Hendrycks, D., & Dietterich, T. (2019). Benchmarking neural network robustness to common corruptions and perturbations. In *International Conference on Learning Representations (ICLR)*.

5. Niculescu-Mizil, A., & Caruana, R. (2005). Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning (ICML)* (pp. 625–632).

6. Ovadia, Y., Fertig, E., Ren, J., Nado, Z., Sculley, D., Nowozin, S., ... & Snoek, J. (2019). Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems (NeurIPS)* (pp. 13991–14002).

7. Platt, J. (1999). Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers* (pp. 61–74).

8. Zadrozny, B., & Elkan, C. (2002). Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 694–699).

9. DeGroot, M. H., & Fienberg, S. E. (1983). The comparison and evaluation of forecasters. *Journal of the Royal Statistical Society: Series D (The Statistician), 32*(1–2), 12–22.

10. Kull, M., Silva Filho, T., & Flach, P. (2017). Beta calibration: A well-founded and easily implemented improvement on logistic calibration for binary classifiers. In *Artificial Intelligence and Statistics (AISTATS)* (pp. 623–631).

11. APPENDICES

11.1 APPENDIX A

Core Model Training Code

Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
from sklearn.calibration import calibration_curve
```

Train-Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Logistic Regression Model

```
lr = LogisticRegression(max_iter=1000)
lr.fit(X_train, y_train)
lr_preds = lr.predict(X_test)
lr_conf = lr.predict_proba(X_test)[:, 1]
```

Random Forest Model

```
rf = RandomForestClassifier(random_state=42)
rf.fit(X_train, y_train)
rf_preds = rf.predict(X_test)
rf_conf = rf.predict_proba(X_test)[:, 1]
```

11.2 APPENDIX B

Reliability Metric Calculations

Logistic Regression Metrics

```
lr_accuracy = accuracy_score(y_test, lr_preds)
lr_avg_conf = lr_conf.mean()
lr_overconfidence = lr_conf[lr_preds != y_test].mean()
lr_conf_gap = lr_avg_conf - lr_accuracy
```

Random Forest Metrics

```
rf_accuracy = accuracy_score(y_test, rf_preds)
rf_avg_conf = rf_conf.mean()
rf_overconfidence = rf_conf[rf_preds != y_test].mean()
rf_conf_gap = rf_avg_conf - rf_accuracy
```

11.3 APPENDIX C

Label Corruption Function

```
def introduce_label_noise(y, noise_level, random_state=42):
    np.random.seed(random_state)
    y_noisy = y.copy()
    n_samples = len(y)
    n_flip = int(noise_level * n_samples)

    flip_indices = np.random.choice(n_samples, n_flip, replace=False)
    y_noisy.iloc[flip_indices] = 1 - y_noisy.iloc[flip_indices]
    return y_noisy
```

11.4 APPENDIX D

Feature Noise Injection

```
def add_feature_noise(X, std_dev):
    noise = np.random.normal(0, std_dev, X.shape)
    return X + noise
```

11.5 APPENDIX E

Missing Data Simulation

```
def introduce_missing_data(X, missing_rate, random_state=42):
    np.random.seed(random_state)
    X_missing = X.copy()
    mask = np.random.rand(*X.shape) < missing_rate
    X_missing[mask] = np.nan
    return X_missing
```

11.6 APPENDIX F

 # Distribution Shift: Age-based split

```
train_shift = data[data["Age"] < 40]
test_shift = data[data["Age"] >= 40]

X_train_shift = train_shift.drop("Target", axis=1)
y_train_shift = train_shift["Target"]

X_test_shift = test_shift.drop("Target", axis=1)
y_test_shift = test_shift["Target"]
```

11.7 APPENDIX G

Calibration Curve

```
prob_true, prob_pred = calibration_curve(y_test, lr_conf, n_bins=10)
plt.plot(prob_pred, prob_true, marker='o', label='Logistic
Regression')
plt.plot([0, 1], [0, 1], linestyle='--', label='Perfect Calibration')
plt.xlabel("Predicted Probability")
plt.ylabel("True Probability")
plt.legend()
plt.show()
```